Modelling and Analysis of a Water Pipeline

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GitHub Repository: Pipeline Optimization

Abstract

This study presents the modelling and optimisation of the vertical alignment of a water pipeline across rolling terrain subject to grade and change-in-grade constraints. The objective is to determine an elevation profile satisfying all design regulations while minimising the total cost of cutting, filling, dumping, and borrowing.

The elevations at Station 1 and Station 15 are fixed at 51 m and 22 m respectively. The maximum grade allowed is 5% and the maximum change in grade is 6%. The boundary grades are 4% before Station 1 and 3% after Station 15.

Trial-and-error solutions were first generated, followed by Linear Programming optimisation using PuLP in Python. The LP model produced the minimum cost solution and also showed the existence of multiple optimal solutions differing only by very small numerical values.

Cost Calculation Method

Width of clearance = 20 m

Distance between stations = 100 m

$$Area = 100 \times 20 = 2000 \text{ m}^2$$

Volume of earthwork

$$V_i = 2000|h_i - x_i|$$

Total cut volume

$$C = \sum V_i^{cut}$$

Total fill volume

$$F = \sum V_i^{fill}$$

Balance

$$C - F = D - B$$

Total cost

$$Z = 150C + 100F + 70D + 120B$$

This formula is used for Trial 1, Trial 2 and Optimal solution.

Trial Solution 1

Station	Elevation	Grade	Change in Grade
1	51	-1	-
2	50	-5	-4
3	45	-5	0
4	40	-5	0
5	35	-5	0
6	30	-1	4
7	29	5	6
8	34	5	0
9	39	1	-4
10	40	-5	-6
11	35	-5	0
12	30	-5	0
13	25	0	5
14	25	-3	-3
15	22	-	-

Total Cost = 50800000

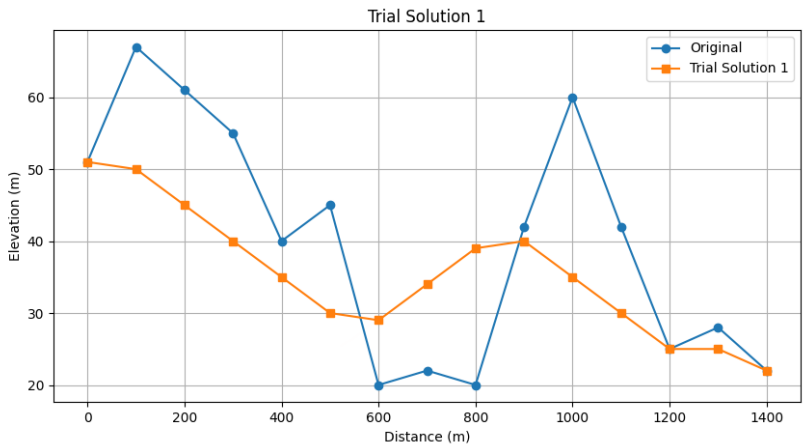

Figure 1: Trial Solution 1

Trial Solution 2

Station	Elevation	Grade	Change in Grade
1	51	2	-
2	53	4	2
3	57	-2	-6
4	55	-5	-3
5	50	-5	0
6	45	-5	0
7	40	-5	0
8	35	-5	0
9	30	1	6
10	31	-5	-6
11	26	1	6
12	27	-2	-3
13	25	0	2
14	25	-3	-3
15	22	-	-

Total Cost = 38820000

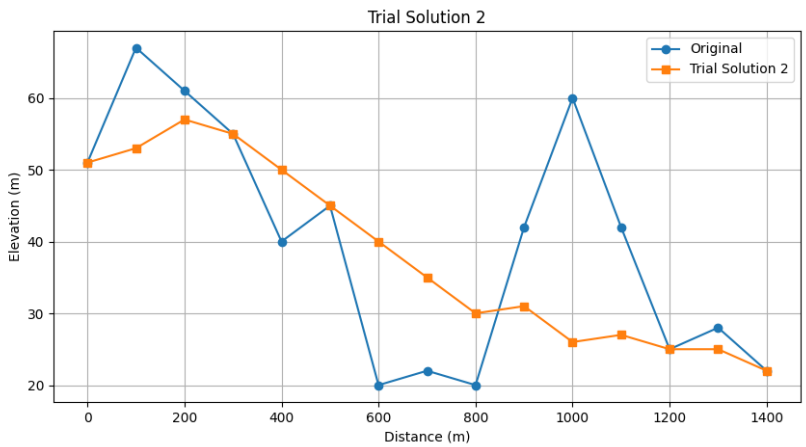

Figure 2: Trial Solution 2

Linear Programming Formulation

$$x_i = \text{elevation at station i}$$

$$x_1 = 51, \quad x_{15} = 22$$

$$x_{i+1} - x_i \leq 5$$

$$x_i - x_{i+1} \leq 5$$

$$x_{i+1} - 2x_i + x_{i-1} \leq 6$$

$$-(x_{i+1} - 2x_i + x_{i-1}) \leq 6$$

$$x_i = h_i - C_i + F_i$$

$$Z = 150 \sum C_i + 100 \sum F_i + 70D + 120B$$

Optimal LP Solution

Station	Elevation	Grade	Change in Grade
1	51	5	-
2	56	0.588	-4.41
3	56.588	-5	-5.58
4	51.588	-5	0
5	46.588	-5	0
6	41.588	-5	0
7	36.588	-4.999	0
8	31.588	0.999	5.99
9	32.588	5	4
10	37.588	2.7	-2.3
11	40.294	-3.294	-6
12	37	-5	-1.71
13	32	-5	0
14	27	-5	0
15	22	-	-

Total Cost = 13088.235

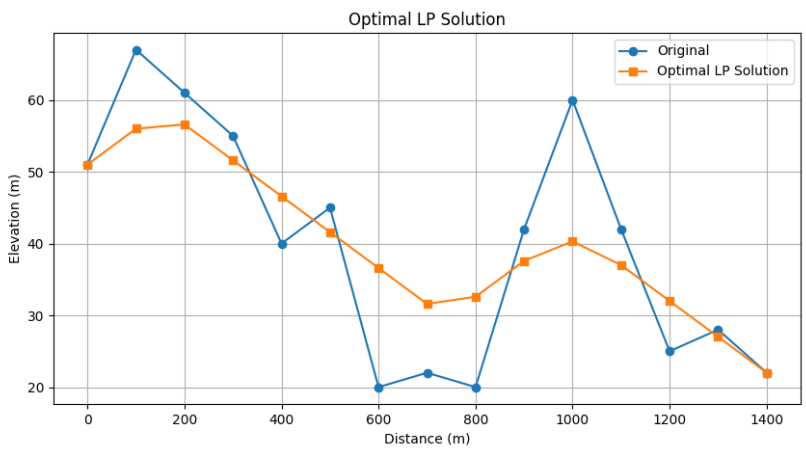

Figure 3: Optimal Solution

Sensitivity Analysis

Question 1

Yes, alternative elevation profiles can give the same total cost as the least-cost solution. Sensitivity analysis of the LP model showed multiple optimal solutions when the objective value was fixed. These solutions differ only by very small numerical changes in elevations (about 10^{-5} m), which do not affect feasibility or cost.

Question 2

Allowing one segment to exceed the 5% limit by 0.5% changes

$$|x_{i+1} - x_i| \leq 5$$

to

$$|x_{i+1} - x_i| \leq 5.5$$

In the optimal table, the grade between Station 8 and Station 9 is exactly 5, so the constraint is binding.

Therefore the best segment to relax is

Station 8 – Station 9

Relaxing this constraint reduces required earthwork, leading to a reduction in cost proportional to the shadow price times 0.5.

Question 3

Arguments in Favour	Arguments Against
More flexibility in grades may reduce earthwork cost	Additional terrain data required
Boundary constraints may be forcing extra cost	Boundary elevations may be fixed
Smoother transitions possible	Larger model complexity
Possible cost reduction	Current solution already optimal in given stretch

Extending the alignment may reduce cost if boundaries are flexible, but if end conditions are fixed, current model is sufficient.